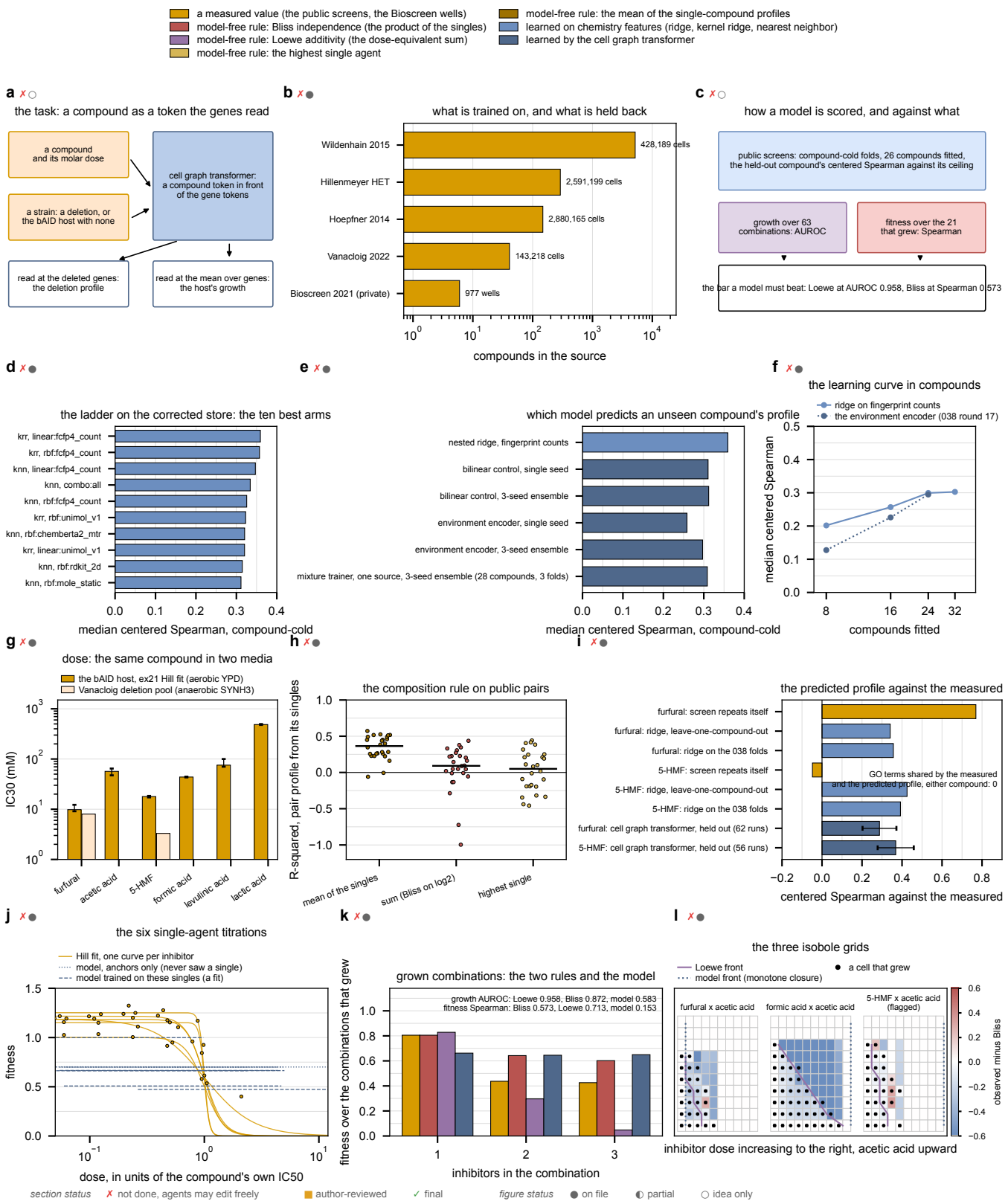

a measured value (the public screens, the Bioscreen wells)
model-free rule: Bliss independence (the product of the singles)
model-free rule: Loewe additivity (the dose-equivalent sum)
model-free rule: the highest single agent
model-free rule: the mean of the single-compound profiles
learned on chemistry features (ridge, kernel ridge, nearest neighbor)
learned by the cell graph transformer
a
the task: a compound as a token the genes read
a compound and its molar dose
a strain: a deletion, or the bAID host with none
cell graph transformer: a compound token in front of the gene tokens
read at the deleted genes: the deletion profile
read at the mean over genes: the host's growth
b
what is trained on, and what is held back
Wildenhain 2015
Hillenmeyer HET
Hoepfner 2014
Vanacloig 2022
Bioscreen 2021 (private)
428,189 cells
2,591,199 cells
2,880,165 cells
143,218 cells
977 wells
compounds in the source
c
how a model is scored, and against what
public screens: compound-cold folds, 26 compounds fitted, the held-out compound's centered Spearman against its ceiling
growth over 63 combinations: AUROC
fitness over the 21 that grew: Spearman
the bar a model must beat: Loewe at AUROC 0.958, Bliss at Spearman 0.573
d
the ladder on the corrected store: the ten best arms
krr, linear:fcfp4_count
krr, rbf:fcfp4_count
knn, linear:fcfp4_count
knn, combo:all
knn, rbf:fcfp4_count
krr, rbf:unimol_v1
knn, rbf:chemberta2_mtr
krr, linear:unimol_v1
knn, rbf:rdkit_2d
knn, rbf:mole_static
median centered Spearman, compound-cold
e
which model predicts an unseen compound's profile
nested ridge, fingerprint counts
bilinear control, single seed
bilinear control, 3-seed ensemble
environment encoder, single seed
environment encoder, 3-seed ensemble
mixture trainer, one source, 3-seed ensemble (28 compounds, 3 folds)
median centered Spearman, compound-cold
f
the learning curve in compounds
ridge on fingerprint counts
the environment encoder (038 round 17)
median centered Spearman
compounds fitted
g
dose: the same compound in two media
the bAID host, ex21 Hill fit (aerobic YPD)
Vanacloig deletion pool (anaerobic SYNH3)
IC30 (mM)
furfural
acetic acid
5-HMF
formic acid
levulinic acid
lactic acid
h
the composition rule on public pairs
R-squared, pair profile from its singles
mean of the singles
sum (Bliss on log2)
highest single
i
the predicted profile against the measured
furfural: screen repeats itself
furfural: ridge, leave-one-compound-out
furfural: ridge on the 038 folds
5-HMF: screen repeats itself
5-HMF: ridge, leave-one-compound-out
5-HMF: ridge on the 038 folds
furfural: cell graph transformer, held out (62 runs)
5-HMF: cell graph transformer, held out (56 runs)
GO terms shared by the measured and the predicted profile, either compound: 0
centered Spearman against the measured
j
the six single-agent titrations
Hill fit, one curve per inhibitor
model, anchors only (never saw a single)
model trained on these singles (a fit)
fitness
dose, in units of the compound's own IC50
k
grown combinations: the two rules and the model
growth AUROC: Loewe 0.958, Bliss 0.872, model 0.583
fitness Spearman: Bliss 0.573, Loewe 0.713, model 0.153
fitness over the combinations that grew
inhibitors in the combination
l
the three isobole grids
Loewe front
model front (monotone closure)
a cell that grew
furfural x acetic acid
formic acid x acetic acid
5-HMF x acetic acid (flagged)
observed minus Bliss
inhibitor dose increasing to the right, acetic acid upward
section status
not done, agents may edit freely
author-reviewed
final
figure status
on file
partial
idea only